



The 2018 Kerala Floods

A State Underwater

In August 2018, Kerala endured its worst flooding in nearly a century — a catastrophe that tested the limits of human resilience, governance, and compassion.

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THE BREAKING POINT

When the Skies Opened

42%

Excess Rainfall

Above-normal monsoon precipitation in a single season

483

Lives Lost

Confirmed fatalities across affected districts

1M+

Displaced

People forced from their homes at the peak of the crisis

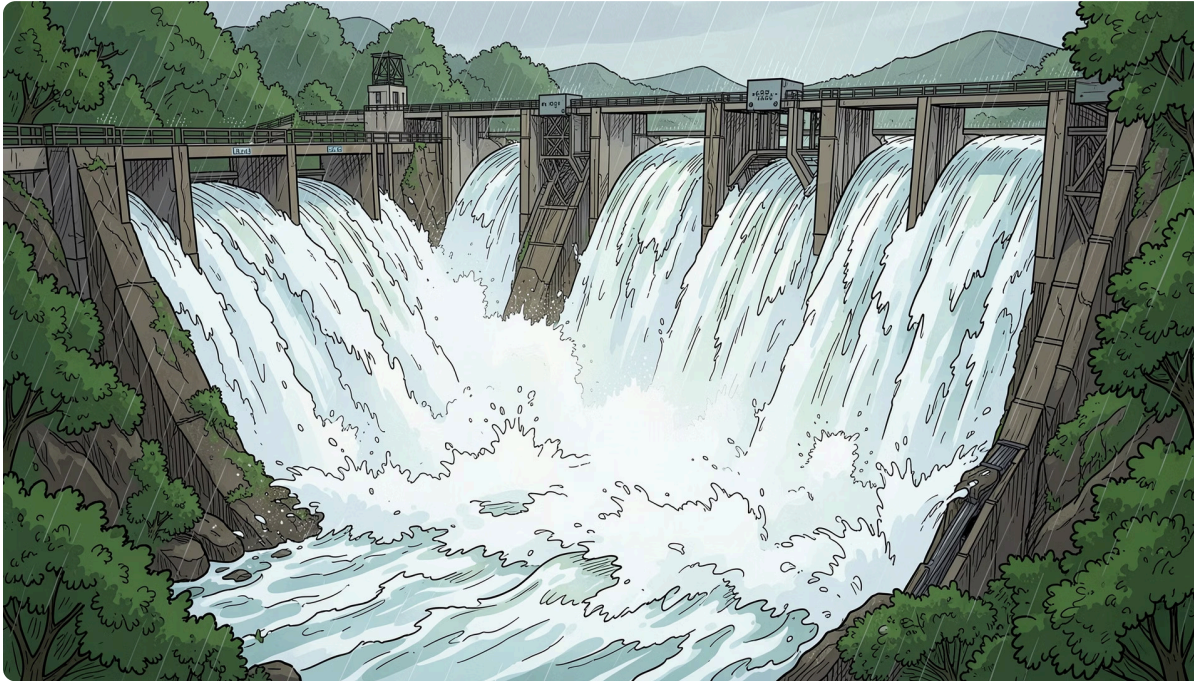
80

Dams at Capacity

Reservoirs overwhelmed simultaneously across the state



Systemic Failures Amplified the Crisis



Heavy precipitation alone did not cause the catastrophe. A cascade of institutional failures turned a natural event into a human disaster.

Sudden Dam Releases

34 out of 42 dams released water simultaneously, with little warning to downstream communities.

Emergency Plan Failures

Emergency action plans were inadequately followed, delaying critical evacuation decisions.

No Coordinated Protocol

Absence of real-time data sharing between dam authorities and disaster agencies.



THE LARGEST RESCUE OPERATION

Mobilising a Nation

Air Assets

40 helicopters and 31 aircraft deployed for aerial rescue and supply drops across cut-off regions.

Ground & Water

500+ boats, 90 NDRF teams, and 182 specialised rescue squads worked around the clock.

Global Support

₹24,067 million raised through public donations from India and the global diaspora.

How Technology Led the Response

Keralarescue.in

An official open-source platform became the digital backbone of the entire rescue operation — coordinating volunteers, tracking requests, and managing relief camps in real time.



1 55,000 Volunteers

Coordinated through open-source platforms across the state and globally.

2 50,000+ Rescue Requests

Logged, prioritised, and actioned through the centralised system.

3 3.5 Lakh Camp Inmates

Managed, fed, and supported across hundreds of relief camps.

LESSONS FROM THE DELUGE

What Kerala Taught the World



Early Warning

Real-time data and dam coordination

Climate Adaptation

Recognise climate change as multiplier

Build Back Better

Smart infrastructure and emergency protocols

The 2018 floods exposed critical gaps in dam management, emergency planning, and climate preparedness. The lessons learned have since informed national disaster policy and infrastructure standards across India.

Building a More Resilient Future

The 2018 Kerala floods were a defining moment — not just for a state, but for how India approaches disaster preparedness. The integration of technology, community action, and institutional reform must become the blueprint for every climate-vulnerable region.

Early Warning

Real-time dam data and automated alerts

Community Power

Open-source platforms and volunteer networks

Climate Action

Treat extreme weather as the new normal



Thank You